Alejandro de la Fuente

Curriculum Vitae

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Profile

PhD researcher at the Barcelona Supercomputing Center (BSC) and the University of Barcelona (UB), specialised in large language models (LLMs) as a research method for the social sciences. I use LLMs to measure political content at scale, validate their output against human coding, and build experiments in which participants interact with LLM-driven agents. I apply these methods to online information diets, political polarization and incivility, combining passive web-tracking data with surveys of the same panelists.

Academic appointments and research experience

2025–present **PhD Researcher**, *Barcelona Supercomputing Center (BSC) & University of Barcelona (UB)*, Barcelona, Spain
Research on political information diets and polarization using web-tracking data and surveys. Development of large-scale text classification pipelines with LLMs on HPC infrastructure (MareNostrum 5).

Feb 2024–Dec 2024 **Research Assistant**, *University of Zurich*, Remote, Switzerland
Swiss National Science Foundation (SNF) project on online incivility. Large-scale data collection, database construction, literature review and statistical analysis in R.

Sep 2023–Apr 2024 **Research Assistant**, *Princeton University*, Remote, USA
Large-scale dataset processing and content classification pipelines for empirical research projects.

2021–2022 **Collaboration Scholarship (Ministry of Education)**, *University of Valladolid*, Valladolid, Spain
Research on electricity market structure and market power in European energy markets.

Feb 2022–May 2022 **Internship**, *Economic and Social Council of Castilla y León (CES)*, Valladolid, Spain
Preparation of economic and social reports on regional economic conditions.

Publications and working papers

Published **The Shift from Browsing to AI Prompting Challenges Web-Tracking Data**. With A. Martinez-Serra. *Discover Data*, 4, 23 (2026). `https://doi.org/10.1007/s44248-026-00122-1`

Preprint **Beyond the Link: Assessing LLMs' Ability to Classify Political Content across Global Media**. With A. Martinez-Serra, N. Visscher, L. Castro and A. S. Cardenal. `https://arxiv.org/abs/2506.17435`

Working paper **Who Reads What—and How Fast? Partisan Asymmetries in Political News Consumption**. With N. Visscher, M. García Ortiz, L. Castro and Y. Zhang. Earlier version: *Bridging the Self-Report Gap: Behavioural Measures of Partisan News Consumption*.

Working paper **Pathways to Polarization: Online Information-Seeking Among Political Extremists**. With L. Castro, Y. Zhang, F. Esser, K. Syed Ali and M. Amsler.

Working paper **Leveraging Large Language Models to Enhance Supervised Classification of Political Incivility on Social Media**. With L. Castro, M. García Ortiz, I. Camacho, A. Casadevall and P. Szewach.

Working paper **What Counts as AI Use? Comparing Self-Reports and Digital Traces in a Spanish Online Panel**. With A. Martinez-Serra, J. C. Niculica, A. Rodilla and A. S. Cardenal.

Working paper **Can Humanized Multi-Agent Systems Simulate Online Political Discussion? Evidence from a Collective Turing Test**. With L. Castro, M. García Ortiz, A. Casadevall, R. Kiddle, P. Szewach and A. Casas.

Working paper **What the Platform Leaves Behind: Iron March's Shutdown and Hate Crime in U.S. Counties**. With M. Pellert.

Conference presentations

2026 BSC Workshop on AI/ML Applied Methods in Computational Social Sciences and Humanities Research, Barcelona: *Who Reads What—and How Fast?*

2026 XVIII Congress of the Spanish Political Science Association (AECPA), Granada: *Who Reads What—and How Fast?* and *Leveraging Large Language Models to Enhance Supervised Classification of Political Incivility*.

2026 European Political Science Society (EPSS), Belfast: *Bridging the Self-Report Gap*.

2026 13th BSC Severo Ochoa Doctoral Symposium, Barcelona: *Pathways to Polarization*.

2025 International Conference on Computational Social Science (IC2S2), Norrköping: *Pathways to Polarization*.

2025 European Political Science Association (EPSA), Madrid: *Pathways to Polarization* and *When the Internet Sheds Light on Corruption*.

2025 International Communication Association (ICA), Denver: *Pathways to Polarization*.

2025 Doctoral seminar, Law and Political Science PhD programme, University of Barcelona: *Scrolling into Extremes: Tracing the Digital Echo Chamber's Road to the Polls*; discussant.

Research projects

Horizon Europe **WHAT-IF**: A Digital Twin of the Information Environment. Grant agreement No. 101177574. PhD researcher; attended the project General Assembly, Amsterdam (January 2025). `https://what-if-horizon.eu/`

AEI (Spain) **DEMOS-AI**: Decoding Moral and Sociopolitical Biases in AI. PID2025-172779NA-I00. Research team. `https://demos-ai.duckdns.org/`

"la Caixa" **CSSINFOYOUTH**: Political information diets of young people in Spain on TikTok, Instagram and YouTube. Social Research call 2024, PI Laia Castro (UB). Research team.

AEI (Spain) **ONLINEHATE**: El efecto de la incivilidad online en la participación electoral: evidencia de cinco redes sociales. PID2023-150282NA-I00. Research team.

SNF Project No. 204483 — From Uncivil Disagreement to Political Unrest? A Cross-Platform and Cross-National Analysis of the Offline Consequences of Online Incivility. Research assistant (2024).

Teaching

2026 Lecturer (with P. Szewach), BSC AI Factory course AI for Social Sciences and Humanities.

Service

2026 Organizing committee, BSC Workshop on AI/ML Applied Methods in Computational Social Sciences and Humanities Research.

2026 Reviewer, *Research & Politics*.

2026 Event volunteer, Dataverse Community Meeting 2026, Barcelona Supercomputing Center.

2025 Organizing committee, 12th BSC Severo Ochoa Doctoral Symposium.

2024 Organizing team, Computational Social Science Conference: innovative methods, research workflows and data stewardship (CODATA, BSC and Fundació la Caixa).

Education

2025–present **PhD**, *Law and Political Science programme, University of Barcelona*, Barcelona, Spain
Doctoral research in computational social science and political communication. Supervisors: Dr Laia Castro (UB) and Dr Max Pellert (BSC).

2022–2024 **MSc, Institutions and Political Economy**, *University of Barcelona*, Barcelona, Spain
Average grade: 9.02/10. Honors in Research Design, Game Theory, and Data Science II.

2018–2022 **BSc, Economics**, *University of Valladolid*, Valladolid, Spain
Average grade: 8.39/10. Multiple honors across core economics courses and honors distinction in final thesis.

Training

2025 ICA Journals Peer Review Workshop for Student and Early-Career Scholars, Denver.

2024 Using Large Language Models for Empirical Research in Social Science (Gaël Le Mens), IBEI Graduate Summer School.

Methods and tools

Programming R; Python; LaTeX; Git.

Methods LLM-based classification and validation; LLM-driven experiments; text-as-data; causal inference with observational data; web-tracking and survey linkage.

Infrastructure HPC (MareNostrum 5); reproducible workflows. Software: STAGELab, turing-test-platform.

Languages Spanish (native), English (B2).